Weighing of 5 objets

Table of Contents

Strategy 1 (offset, and then one by one) 1
Matrix of experiments E 1
Bar chart 1
Analysis of strategy 1 2
Strategy 2 (all, then one by one) 2
Bar chart 3
Analysis of strategy 2 4
Strategy 4 (Offset, then two by two) 4
Matrix of experiments 4
Analysis of strategy 4 5
Strategy 5 (All, then two by two) 5
Analysis strategy 5 6
Strategy 6 (Offset, then three by three) 7
Analysis of strategy 6 8
Strategy 7 (All, then three by three) 8
Analysis of strategy 7 9
Strategy 8 (5 columns of an Hadamard matrix) 9
Analysis of strategy 8 10
Balance 10
How evolves the trace? 10
How evolve the determinant ? 11
C 12
Fonction d'analyse 12


```
Axe1=[0 7 0 4];
```

Strategy 1 (offset, and then one by one)

Matrix of experiments E

```
X1=[0 0 0 0 0;eye(5)];

disp(X1)
```

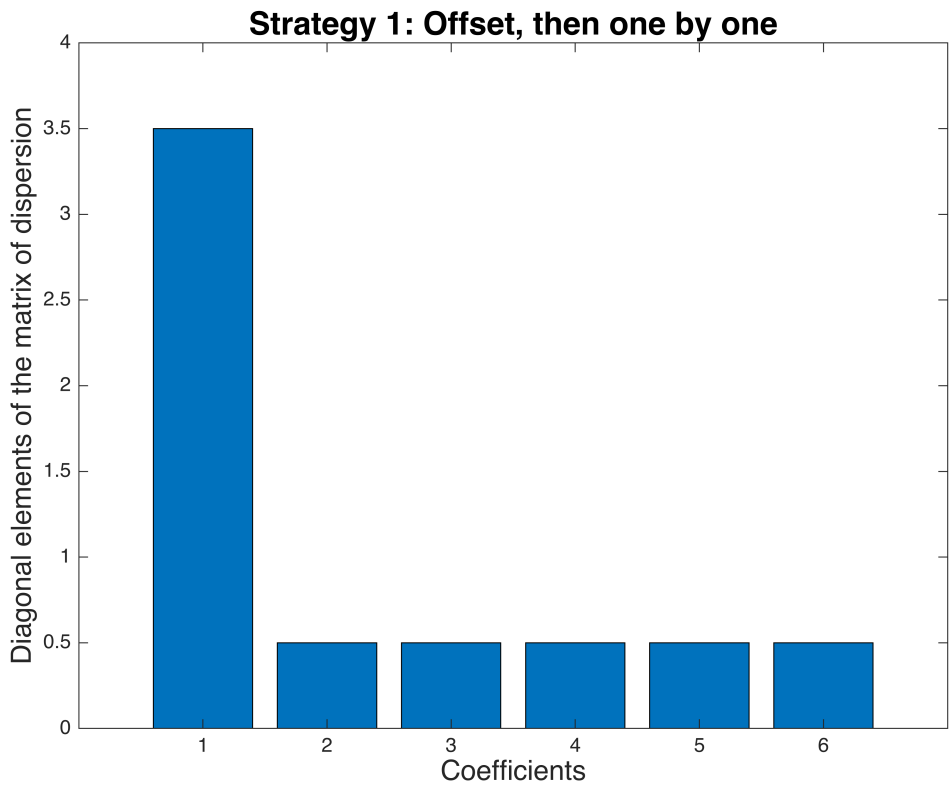
```
     0     0     0     0     0
     1     0     0     0     0
     0     1     0     0     0
     0     0     1     0     0
     0     0     0     1     0
     0     0     0     0     1
```

Bar chart

```
titre='Strategy 1: Offset, then one by one';
```

```
[T(1),DET(1)]=analyse(X1,Axe1,titre);
```

```
Standardization of the matrix of experiment E

    -1    -1    -1    -1    -1
     1    -1    -1    -1    -1
    -1     1    -1    -1    -1
    -1    -1     1    -1    -1
    -1    -1    -1     1    -1
    -1    -1    -1    -1     1

Matrix of dispersion D

    3.5000    1.0000    1.0000    1.0000    1.0000    1.0000
    1.0000    0.5000    0.2500    0.2500    0.2500    0.2500
    1.0000    0.2500    0.5000    0.2500    0.2500    0.2500
    1.0000    0.2500    0.2500    0.5000    0.2500    0.2500
    1.0000    0.2500    0.2500    0.2500    0.5000    0.2500
    1.0000    0.2500    0.2500    0.2500    0.2500    0.5000
```



Analysis of strategy 1

1. We observe that the variance on the offset a_o is multiplied by 3.5 in comparison with the experimental variance.
2. The viariance of the other coefficients of the model is equal to half of the experimental variance, whereas the optimun is 1/6.

Strategy 2 (all, then one by one)

Matrix of experiments

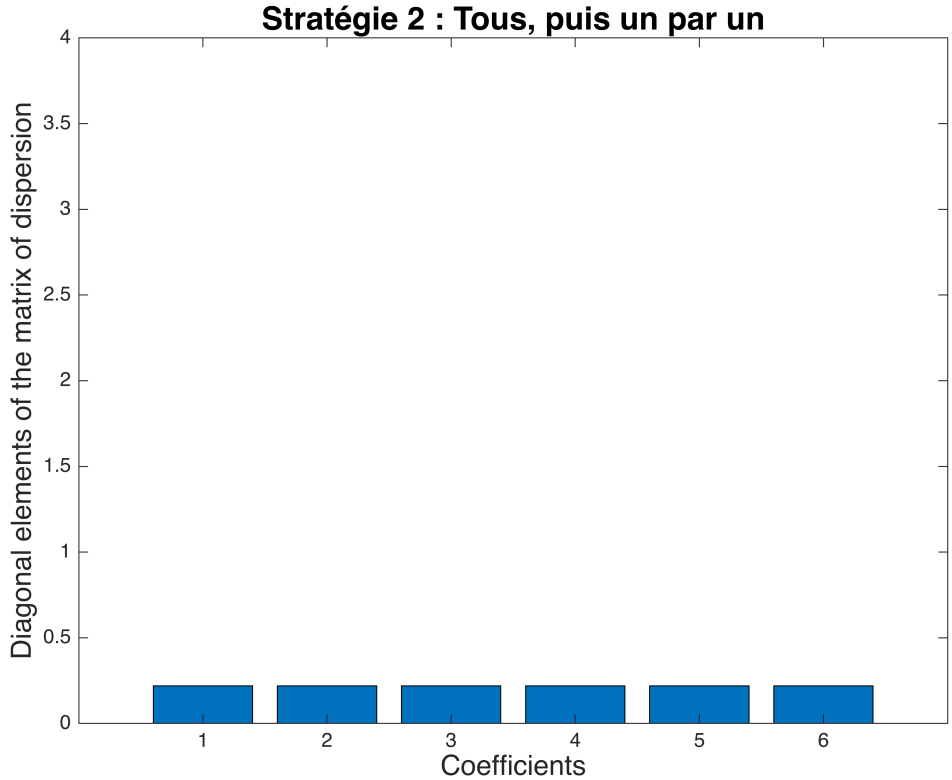
```
X2=[1 1 1 1 1;eye(5)];

disp(X2)
```

```
     1     1     1     1     1
     1     0     0     0     0
     0     1     0     0     0
     0     0     1     0     0
     0     0     0     1     0
     0     0     0     0     1
```

Bar chart

```
titre='Stratégie 2 : Tous, puis un par un';
[T(2),DET(2)]=analyse(X2,Axe1,titre);
```

```
Standardization of the matrix of experiment E

     1     1     1     1     1
     1    -1    -1    -1    -1
    -1     1    -1    -1    -1
    -1    -1     1    -1    -1
    -1    -1    -1     1    -1
    -1    -1    -1    -1     1

Matrix of dispersion D

    0.2188    0.0313    0.0313    0.0312    0.0312    0.0312
    0.0313    0.2188   -0.0313   -0.0313   -0.0312   -0.0312
    0.0313   -0.0313    0.2188   -0.0313   -0.0313   -0.0312
    0.0312   -0.0313   -0.0313    0.2188   -0.0312   -0.0313
    0.0312   -0.0312   -0.0313   -0.0312    0.2188   -0.0313
    0.0312   -0.0312   -0.0312   -0.0313   -0.0313    0.2188
```



Analysis of strategy 2

1. The variance of the model coefficients corresponds to 22% of the experimental variance, while the optimun is 17% (1/6).

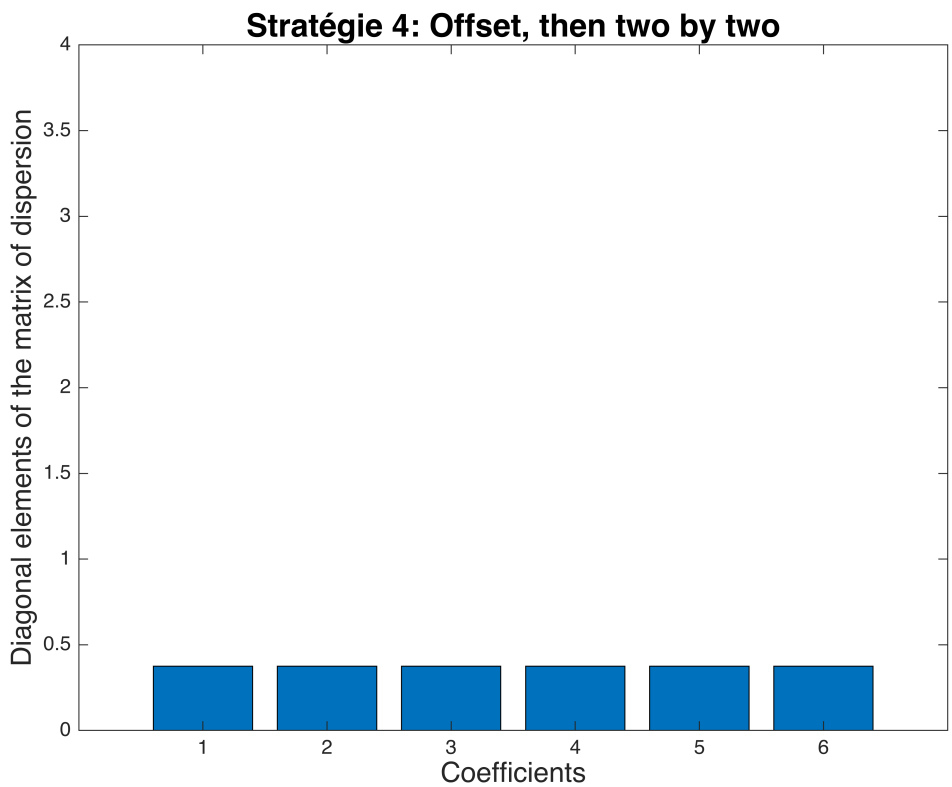
Strategy 4 (Offset, then two by two)

Matrix of experiments

```
X3=[0 0 0 0 0
   1 1 0 0 0
   0 1 1 0 0
   0 0 1 1 0
   0 0 0 1 1
```

```
   1 0 0 0 1];
titre='Stratégie 4: Offset, then two by two';
[T(3),DET(3)]=analyse(X3,Axe1,titre);
```

```
Standardization of the matrix of experiment E

    -1    -1    -1    -1    -1
     1     1    -1    -1    -1
    -1     1     1    -1    -1
    -1    -1     1     1    -1
    -1    -1    -1     1     1
     1    -1    -1    -1     1

Matrix of dispersion D

    0.3750    0.1250    0.1250    0.1250    0.1250    0.1250
    0.1250    0.3750   -0.1250    0.1250    0.1250   -0.1250
    0.1250   -0.1250    0.3750   -0.1250    0.1250    0.1250
    0.1250    0.1250   -0.1250    0.3750   -0.1250    0.1250
    0.1250    0.1250    0.1250   -0.1250    0.3750   -0.1250
    0.1250   -0.1250    0.1250    0.1250   -0.1250    0.3750
```



Analysis of strategy 4

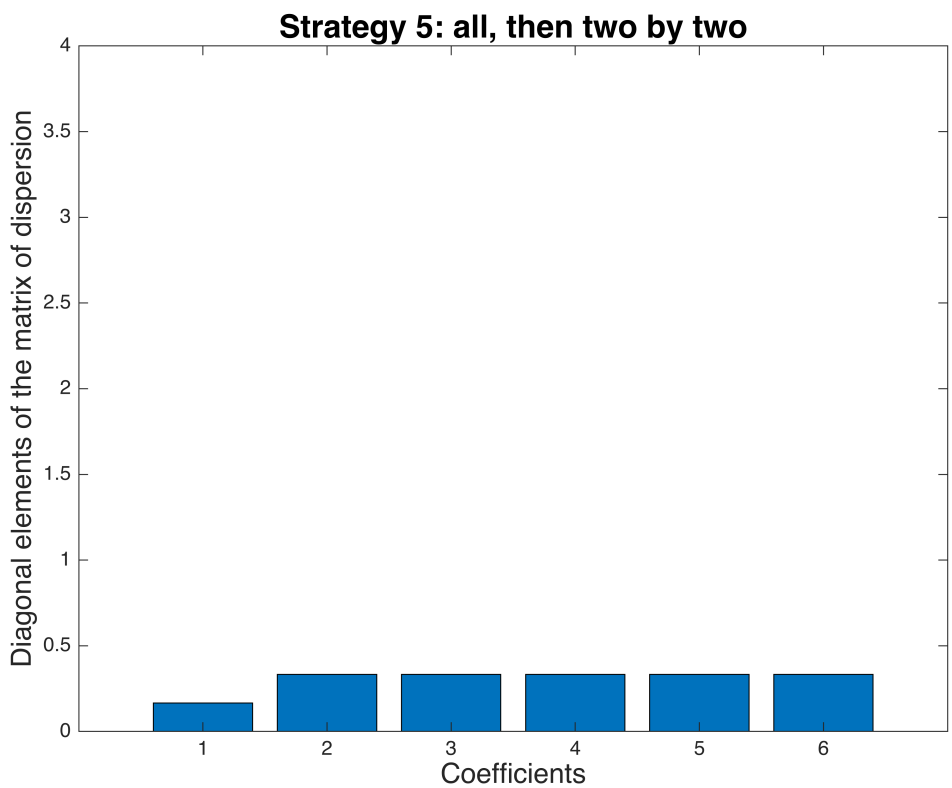
1. The variance of the model coefficients corresponds to 22% of the experimental variance, while the optimun is 17% (1/6).

Strategy 5 (All, then two by two)

```
X4=[1 1 1 1 1
   1 1 0 0 0
```

```
   0 1 1 0 0
   0 0 1 1 0
   0 0 0 1 1
   1 0 0 0 1];

titre='Strategy 5: all, then two by two';

[T(4),DET(4)]=analyse(X4,Axe1,titre);
```

```
Standardization of the matrix of experiment E

     1     1     1     1     1
     1     1    -1    -1    -1
    -1     1     1    -1    -1
    -1    -1     1     1    -1
    -1    -1    -1     1     1
     1    -1    -1    -1     1

Matrix of dispersion D

    0.1667         0         0         0         0         0
         0    0.3333   -0.1667    0.0833    0.0833   -0.1667
         0   -0.1667    0.3333   -0.1667    0.0833    0.0833
         0    0.0833   -0.1667    0.3333   -0.1667    0.0833
         0    0.0833    0.0833   -0.1667    0.3333   -0.1667
         0   -0.1667    0.0833    0.0833   -0.1667    0.3333
```



Analysis strategy 5

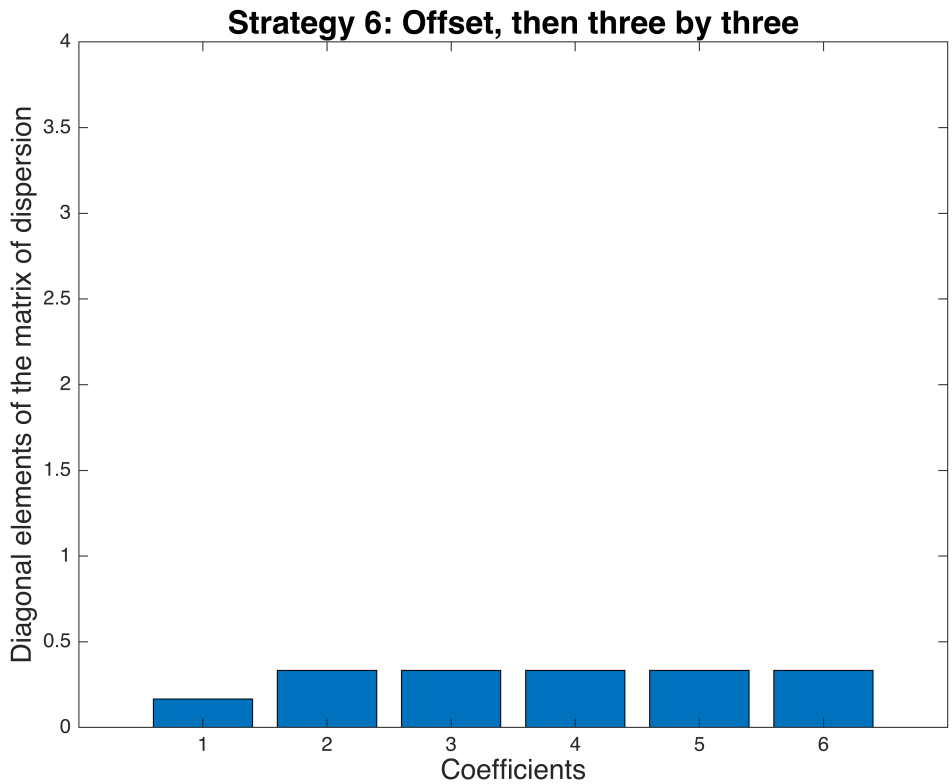
1. The variance of the constant corresponds to 17% of the experimental variance, this is the optimum.

2. The variance of the other coefficients corresponds to 33% of the experimental variance, which is far from the optimum of 17%.

Strategy 6 (Offset, then three by three)

```
X5=[0 0 0 0 0
   1 1 1 0 0
   0 1 1 1 0
   0 0 1 1 1
   1 0 0 1 1
   1 1 0 0 1];
titre='Strategy 6: Offset, then three by three';
[T(5),DET(5)]=analyse(X5,Axe1,titre);
```

```
Standardization of the matrix of experiment E

    -1    -1    -1    -1    -1
     1     1     1    -1    -1
    -1     1     1     1    -1
    -1    -1     1     1     1
     1    -1    -1     1     1
     1     1    -1    -1     1

Matrix of dispersion D

    0.1667         0         0         0         0         0
         0    0.3333   -0.1667    0.0833    0.0833   -0.1667
         0   -0.1667    0.3333   -0.1667    0.0833    0.0833
         0    0.0833   -0.1667    0.3333   -0.1667    0.0833
         0    0.0833    0.0833   -0.1667    0.3333   -0.1667
         0   -0.1667    0.0833    0.0833   -0.1667    0.3333
```



Analysis of strategy 6

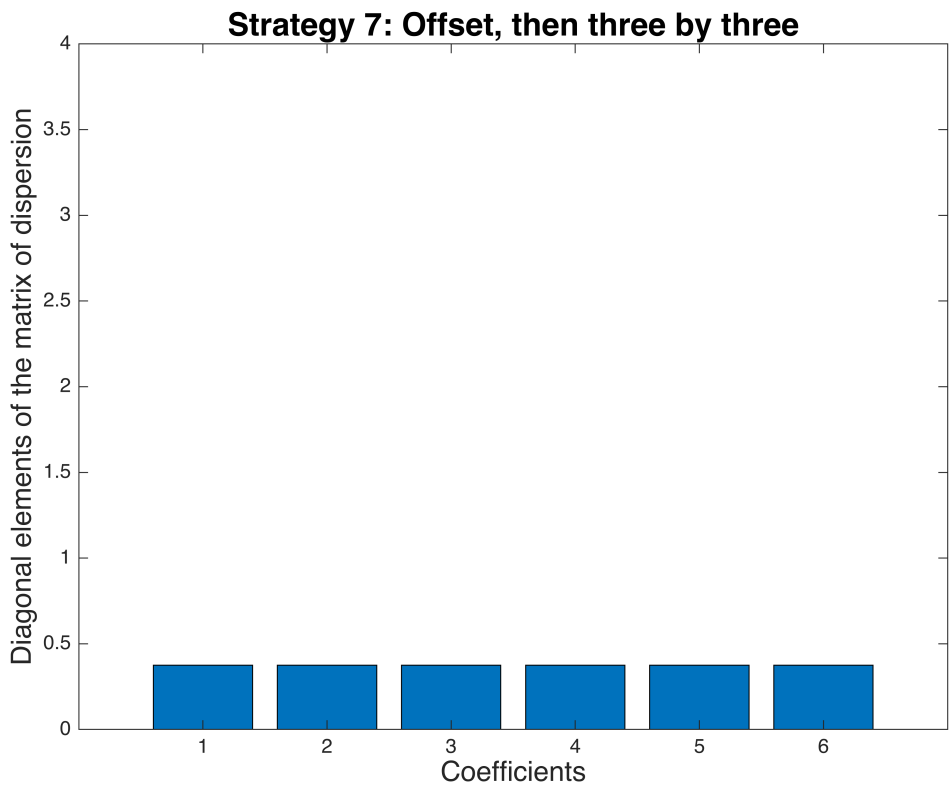
1. Idem stretegy 4.

Strategy 7 (All, then three by three)

```
X7=[1 1 1 1 1
   1 1 1 0 0
   0 1 1 1 0
   0 0 1 1 1
   1 0 0 1 1
   1 1 0 0 1];

titre='Strategy 7: Offset, then three by three';

% call of the function analyse
[T(7),DET(7)]=analyse(X7,Axe1,titre);
```

```
Standardization of the matrix of experiment E

     1     1     1     1     1
     1     1     1    -1    -1
    -1     1     1     1    -1
    -1    -1     1     1     1
     1    -1    -1     1     1
     1     1    -1    -1     1
```

```
Matrix of dispersion D

    0.3750   -0.1250   -0.1250   -0.1250   -0.1250   -0.1250
   -0.1250    0.3750   -0.1250    0.1250    0.1250   -0.1250
   -0.1250   -0.1250    0.3750   -0.1250    0.1250    0.1250
   -0.1250    0.1250   -0.1250    0.3750   -0.1250    0.1250
   -0.1250    0.1250    0.1250   -0.1250    0.3750   -0.1250
   -0.1250   -0.1250    0.1250    0.1250   -0.1250    0.3750
```



Analysis of strategy 7

1. The variance of the model coefficients corresponds to 37.5% of the experimental variance, while the optimun is 1/6,17%

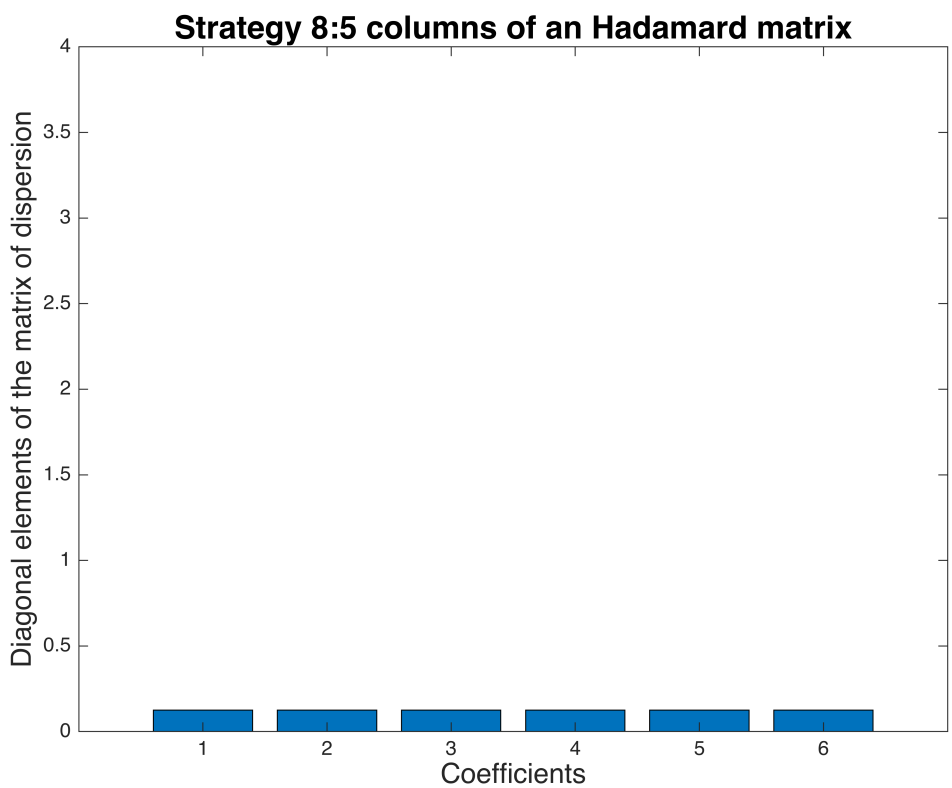
Strategy 8 (5 columns of an Hadamard matrix)

In this strategy we make use of the Hadamard matrix. As it does not exist an Hadamard matrix for 5 factors, we generate the matrix for 8 factors and use only the 6 first columns.

```
% create a matrix of Hadamard 8x8
X8=hadamard(8);

% select the columns 2 to 6 for coding the factors
X8=X8(:,2:6);

titre='Strategy 8:5 columns of an Hadamard matrix';
```

```
% call of the function analyse
[T(8),DET(8)]=analyse(X8,Axe1,titre);
```

```
Standardization of the matrix of experiment E

     1     1     1     1     1
    -1     1    -1     1    -1
     1    -1    -1     1     1
    -1    -1     1     1    -1
     1     1     1    -1    -1
    -1     1    -1    -1     1
     1    -1    -1    -1    -1
    -1    -1     1    -1     1

Matrix of dispersion D

    0.1250         0         0         0         0         0
         0    0.1250         0         0         0         0
         0         0    0.1250         0         0         0
         0         0         0    0.1250         0         0
         0         0         0         0    0.1250         0
         0         0         0         0         0    0.1250
```



Analysis of strategy 8

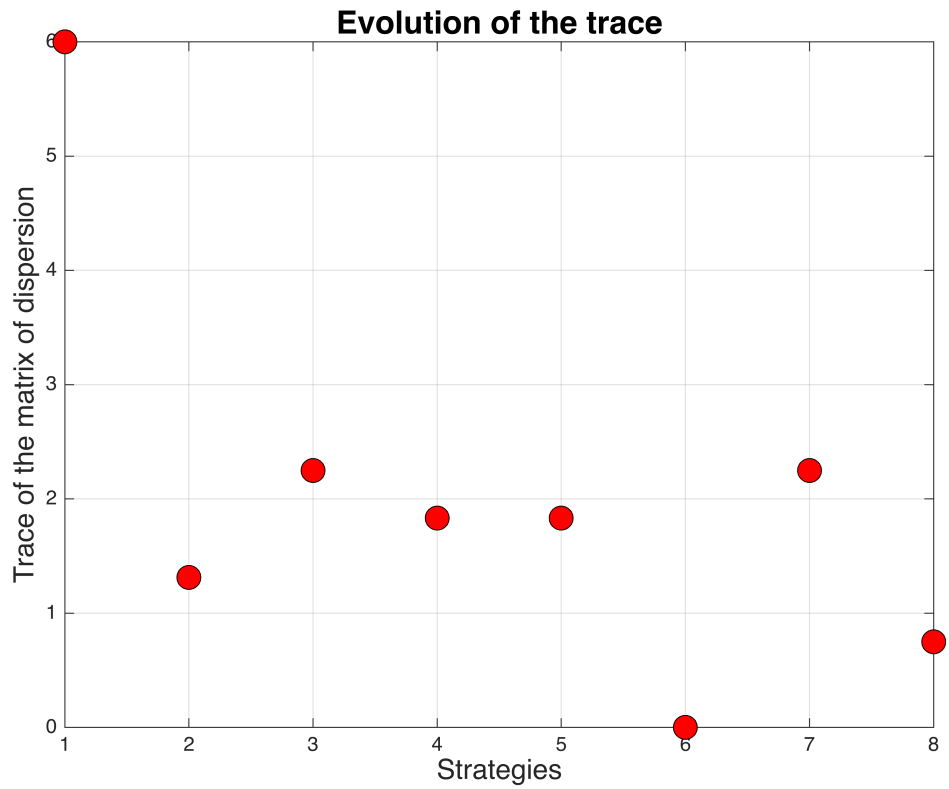
Optimal design: variance of the coefficients corresponds to 1/8 of the experimental variance for 8 experiments.

Balance

How evolves the trace?

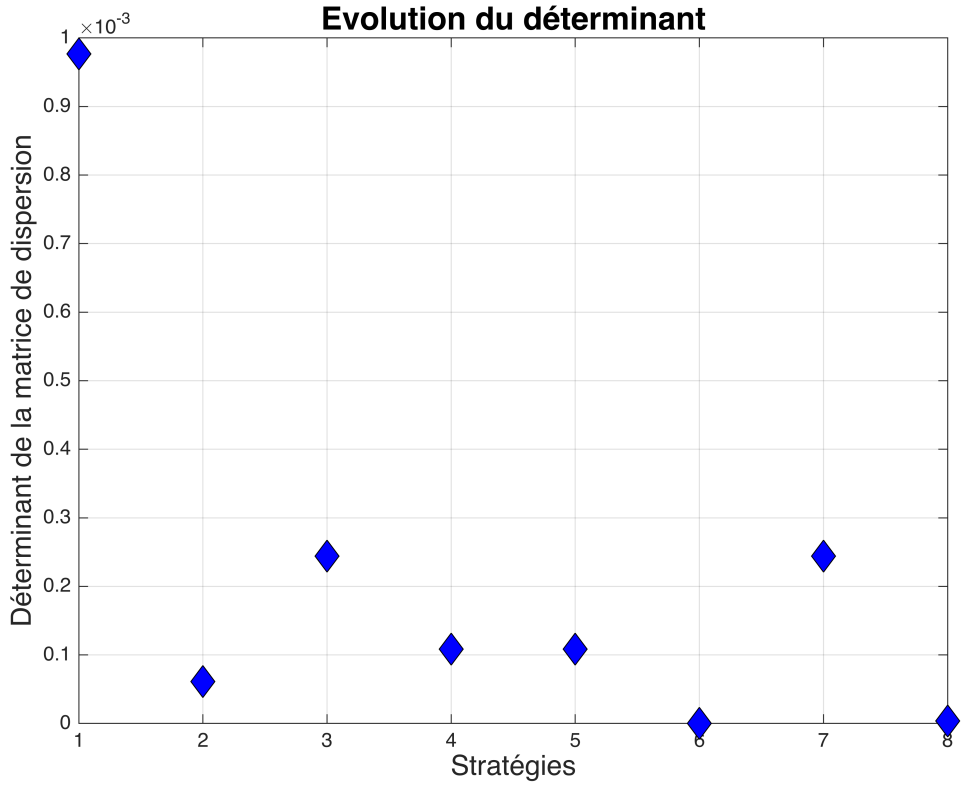
```
%figure
```

```
plot(T,'ok','MarkerSize',12,'MarkerFaceColor','r')
title('Evolution of the trace', 'FontSize',16)
ylabel('Trace of the matrix of dispersion','FontSize',14)
xlabel('Strategies','FontSize',14)
grid on
```



How evolve the determinant ?

```
plot(DET,'dk','MarkerSize',12,'MarkerFaceColor','b')
title('Evolution du déterminant', 'FontSize',16)
ylabel('Déterminant de la matrice de dispersion','FontSize',14)
xlabel('Stratégies','FontSize',14)
grid on
```



Conclusion

1. Les stratégies 2 et 7 sont les meilleures pour 6 expériences.
2. La stratégie 9 est optimale avec 8 expériences.

Fonction d'analyse

```
function [T,DET]=analyse(X,Axe1,titre_string)

% Standardization of he matrix of experiment
X=rescale(X,-1,1,"InputMin",min(X),"InputMax",max(X));
disp('Standardization of the matrix of experiment E ')
disp(' ')
disp(X)

% Compute the model matrix
X=x2fx(X,'linear');

% Compute the matrix of dispersion
D=inv(X'*X);
disp('Matrix of dispersion D')
disp(' ')
```

```
disp(D)

% barchart of the diagonal elements

bar(diag(D))
title(titre_string, 'FontSize',16)
ylabel('Diagonal elements of the matrix of dispersion','FontSize',14)
xlabel('Coefficients','FontSize',14)
axis(Axe1);

% Compute the trace and the déterminant of the dispersion matrix
T=trace(D);
DET=det(D);

end
```